

Term 1 Task 2 2014

MATHEMATICS EXTENSION 2

General Instructions:

- Reading Time: 5 minutes.
- Working Time: 2 hours.
- Write in black pen.
- Board approved calculators & templates may be used
- A Standard Integral Sheet is provided.
- In Question 6 - 9, show all relevant mathematical reasoning and/or calculations.
- Marks may not be awarded for careless or badly arranged working.

Total Marks 85

Section I: 5 marks

- Attempt Question 1 – 5.
- Answer on the Multiple Choice answer sheet provided.
- Allow about 8 minutes for this section.

Section II: 80 Marks

- Attempt Question 6 - 9
- Answer on paper provided unless otherwise instructed. Start a new page for each new question.
- Allow about 1 hours & 42 minutes for this section.

The answers to all questions are to be returned in separate *stapled* bundles clearly labelled Question 6, Question 7, etc. Each question must show your Candidate Number.

For the following questions colour the most correct answer on your multiple choice answer sheet

- 1 Which of the following are the foci for the ellipse

$$\frac{x^2}{9} + \frac{y^2}{16} = 1$$

(A) $(0, \pm \frac{3\sqrt{7}}{4})$

(B) $(\pm\sqrt{7}, 0)$

(C) $(0, \pm\sqrt{7})$

(D) $(\pm \frac{3\sqrt{7}}{4}, 0)$

- 2 Using implicit differentiation on $y^3 = x^2 + xy$ then $\frac{dy}{dx} =$

(A) $\frac{3y^2 - 2x}{x}$

(B) $\frac{2x + y}{3y^2 - x}$

(C) $\frac{2x}{3y^2 + y}$

(D) $\frac{2x - y}{3y^2 + y}$

- 3 Using a suitable substitution what is the correct expression for $\int_0^{\frac{\pi}{3}} \sin^3 x \cos^4 x dx$

(A) $\int_0^{\frac{\sqrt{3}}{2}} (u^4 - u^6) du$

(B) $\int_{\frac{1}{2}}^{\frac{1}{2}} (u^6 - u^4) du$

(C) $\int_{\frac{1}{2}}^1 (u^6 - u^4) du$

(D) $\int_0^{\frac{\sqrt{3}}{2}} (u^6 - u^4) du$

4

The points $P(a \cos \theta, b \sin \theta)$ and $Q(a \cos \phi, b \sin \phi)$ lie on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ and the chord PQ subtends a right angle at $(0,0)$. Which of the following is the correct expression?

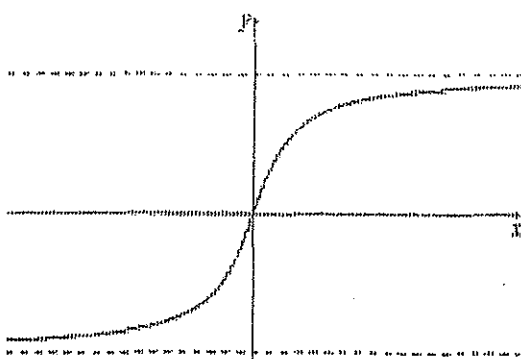
(A) $\tan \theta \tan \phi = -\frac{b^2}{a^2}$

(B) $\tan \theta \tan \phi = \frac{a^2}{b^2}$

(C) $\tan \theta \tan \phi = \frac{b^2}{a^2}$

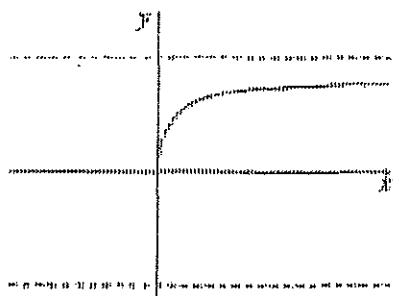
(D) $\tan \theta \tan \phi = -\frac{a^2}{b^2}$

5 The graph of $y = f(x)$ is shown below

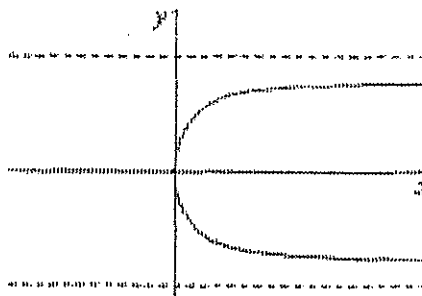


Which of the following best represents $y^2 = f(x)$

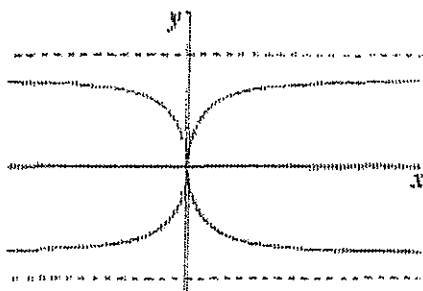
(A)



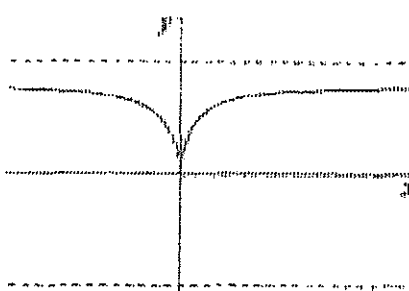
(B)



(C)



(D)



QUESTION 6 (20 Marks)

MARKS

(a) Find $\int \frac{dx}{\sqrt{6x - x^2}}$ 2

(b) Find $\int \frac{1-x}{1-\sqrt{x}} dx$ 2

(c) (i) Show that $\int_0^1 \frac{5-5x^2}{(1+2x)(1+x^2)} dx = \frac{1}{2}(\pi + \ln \frac{27}{16})$ 4

(ii) Hence find $\int_0^{\frac{\pi}{2}} \frac{\cos x}{(1+2\sin x + \cos x)} dx$ using the substitution $t = \tan \frac{x}{2}$ 3

(d) Find $\int_1^3 x^2 \ln x dx$ 4

(e) (i) On the same set of axes sketch and label $y = x^{\frac{1}{3}}$ and $y = e^{-x}$ 2

(ii) Hence, on a different set of axes, without using calculus, sketch and label clearly the graph of the function $y = x^{\frac{1}{3}}e^{-x}$ 2

(iii) Use your sketch to determine the values of m for which the equation $x^{\frac{1}{3}}e^{-x} = mx - 1$ has exactly one solution 1

QUESTION 7 (20 Marks) Start a new page

MARKS

(a) Let P be the point (x_1, y_1) on the ellipse $E: \frac{x^2}{4} + \frac{y^2}{3} = 1$ with $x_1 > 0$.

(i) Find the eccentricity of the ellipse.

1

(ii) Prove that the equation of the tangent at P is $\frac{xx_1}{4} + \frac{yy_1}{3} = 1$.

4

(iii) Find the coordinates of the point T where the tangent meets the nearest directrix

2

(iv) Prove that the segment of the tangent to the ellipse E between the point of contact and the directrix subtends a right angle to the corresponding focus.

3

(v) Write down the equation of the chord of contact PQ from $R(x_0, y_0)$ to E

1

(vi) Show that if PQ passes through the focus then R lies on the directrix.

2

(b) Show that if $y = px + q$ is a tangent to the hyperbola

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \text{ then } p^2 a^2 - b^2 = q^2$$

3

(c) Consider the complex number $z = x + iy$ where $z^2 = a + ib$

(i) Sketch on the same set of axes the graphs of $x^2 - y^2 = a$ and $2xy = b$ where both a and b are positive

The foci and directrices of the curves need not be found

2

(ii) Use the graphs to explain why there are two distinct square roots of the complex number $a + ib$ if $a > 0, b > 0$

1

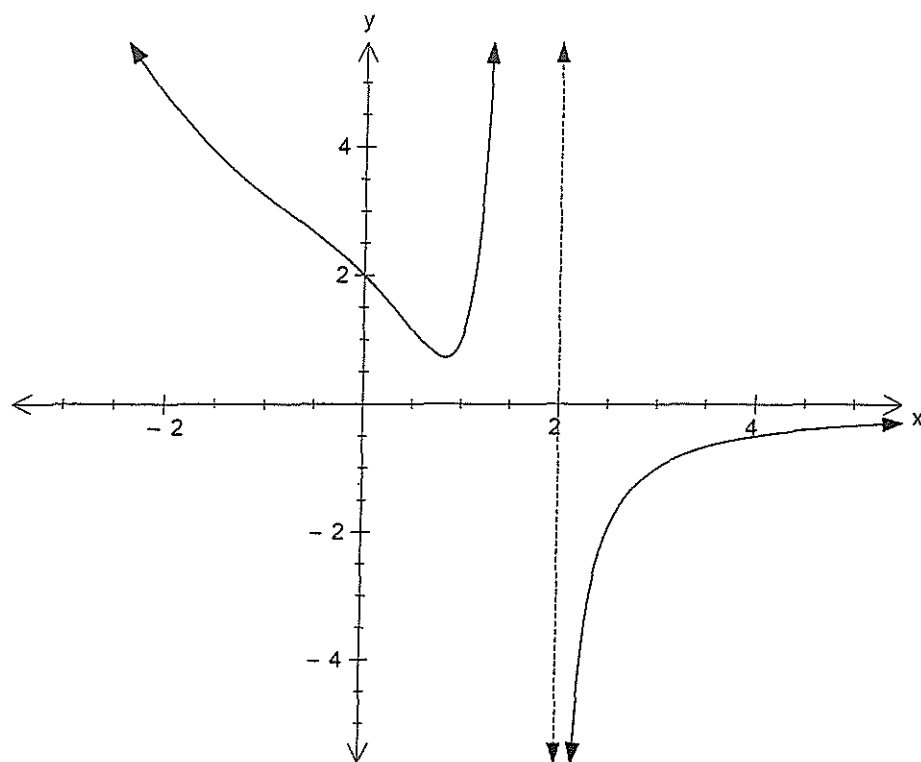
(iii) Consider how the sketch changes when b is negative. What is the relationship between the new square roots and those found when b is positive?

1

QUESTION 8 (20 Marks) Start a new page

MARKS

(a) The diagram below shows the graph of $y = f(x)$



Draw separate sketches of the following (indicate important features)

(i) $y = \frac{1}{0.5 - f(x)}$ 3

(ii) $y = [f(x)]^2$ 2

(iii) $y = f(|x|)$ 2

(iv) $y = \int_{-2}^x f(u) du$ for $-2 \leq x \leq 1$ 2

(b) (i) Show that $(1 - \sqrt{x})^{n-1} \sqrt{x} = (1 - \sqrt{x})^{n-1} - (1 - \sqrt{x})^n$ 2

(ii) If $I_n = \int_0^1 (1 - \sqrt{x})^n dx$ for $n \geq 0$ show that $I_n = \frac{n}{n+2} I_{n-1}$ for $n \geq 1$ 3

(iii) Hence find I_3 2

QUESTION 8 (cont)

(c) Find $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin^5 \theta \, d\theta$

1

(d) Prove that for all complex numbers z and w

(i) $|z + w|^2 + |z - w|^2 = 2(|z|^2 + |w|^2)$

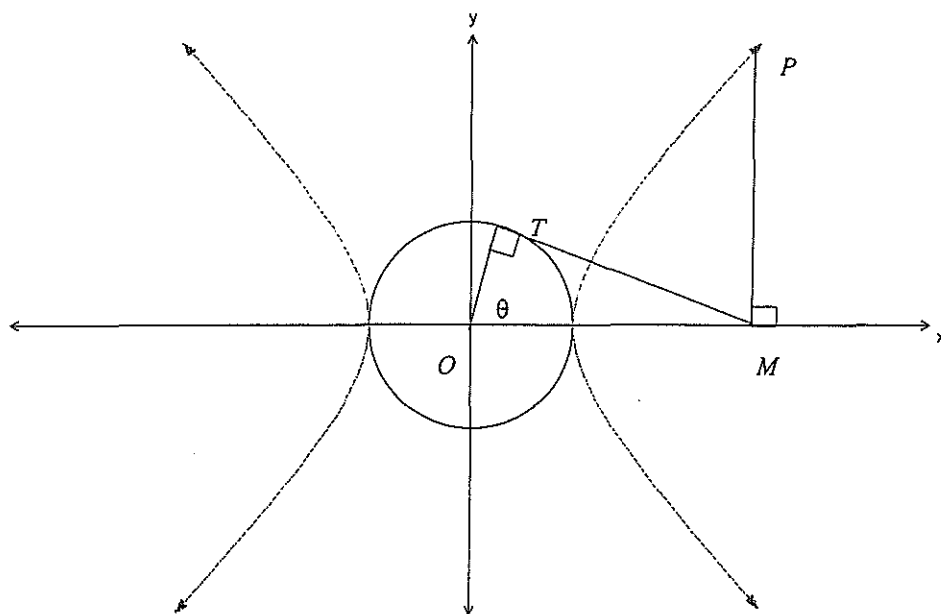
2

(ii) Give a geometrical interpretation of this equation in the complex plane

1

QUESTION 9 (20 Marks) Start a new page

MARKS



(a) The sketch shows the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ and the circle $x^2 + y^2 = a^2$

with $a, b \geq 0$. T lies on the circle where $\angle TOX = \theta$ and $0 \leq \theta \leq \frac{\pi}{2}$. The tangent at T meets OX at M and MP is perpendicular to OX with P on the hyperbola.

i) Find the equation of the tangent TM and hence the coordinates of M.

3

ii) Hence show that the coordinates of P are $(a \sec \theta, b \tan \theta)$

1

QUESTION 9 (cont)

- iii) Assume $\theta \neq \frac{\pi}{4}$, if $Q(a \sec \beta, b \tan \beta)$ is another point on the hyperbola, where $\theta + \beta = \frac{\pi}{2}$ show that the equation of PQ is $ay = b(\cos \theta + \sin \theta)x - ab$. 3

- iv) Every such chord PQ passes through a fixed point. Find the coordinates of this fixed point.. 1

- v) Show that as θ approaches $\frac{\pi}{2}$ the chord PQ approaches a line parallel to an asymptote of the hyperbola 2

- (b) (i) Consider $f(x) = \frac{1}{1 + \tan x}$ where $0 \leq x \leq \frac{\pi}{2}$ and $f(\frac{\pi}{2}) = 0$ 2

Show that $f(x) + f(\frac{\pi}{2} - x) = 1$

- (ii) Hence or otherwise evaluate $\int_0^{\frac{\pi}{2}} \frac{1}{1 + \tan x} dx$ 3

- (c) Using the fact that $\tan^{-1} x + \tan^{-1} y = \tan^{-1} \left(\frac{x + y}{1 - xy} \right)$ for $0 \leq x \leq 1$ and $0 \leq y \leq 1$. Prove by mathematical induction that for all positive integers n 5
- $$\tan^{-1} \left(\frac{1}{2 \times 1^2} \right) + \tan^{-1} \left(\frac{1}{2 \times 2^2} \right) + \dots + \tan^{-1} \left(\frac{1}{2 \times n^2} \right) = \frac{\pi}{4} - \tan^{-1} \left(\frac{1}{2n + 1} \right)$$

END OF EXAM

MATHEMATICS Extension 2: Question 6...

Suggested Solutions	Marks	Marker's Comments
(a) $\int \frac{dx}{\sqrt{9-(x-3)^2}}$ $= \sin^{-1} \left(\frac{x-3}{3} \right) + C$	① ①	Multiple Choice Answers Q1 C Q2 B Q3 B Q4 D Q5 B
(b) $\int \frac{1-x}{1-\sqrt{x}} dx$ $= \int \frac{(1-\sqrt{x})(1+\sqrt{x})}{1-\sqrt{x}} dx$ $= \int (1+\sqrt{x}) dx$ $= x + \frac{2}{3}x^{3/2} + C$	① ①	
(c) (i) $\int_0^1 \frac{5-5x^2}{(1+2x)(1+x^2)} dx$ $= \int_0^1 \left(\frac{a}{1+2x} + \frac{bx+c}{1+x^2} \right) dx$ $\therefore 5-5x^2 = a(1+x^2) + (bx+c)(1+2x)$ pt in $x=0$: $5=a+c$ pt in $x=\frac{1}{2}$: $5-\frac{5}{4}=a(\frac{3}{4})$ $a=3$ $\therefore c=2$ pt in $x=1$: $0=6+6+3b$ $b=-4$ $\therefore I = \int_0^1 \left(\frac{3}{1+2x} - \frac{4x+2}{1+x^2} \right) dx$ $= \int_0^1 \left(\frac{3}{1+2x} - \frac{4x}{1+x^2} + \frac{2}{1+x^2} \right) dx$ $= \left[\frac{3}{2} \ln(1+2x) - 2 \ln(1+x^2) + 2 \tan^{-1} x \right]_0^1$ $= \left(\frac{3}{2} \ln 3 + 2 \tan^{-1} 1 \right) - \left(\frac{3}{2} \ln 1 + 2 \tan^{-1} 0 \right) = (2 \ln 2)$ $= \frac{3}{2} \ln 3 + \pi/2 - 2 \ln 2$ $= \frac{1}{2} (\ln 27 + \pi - \ln 4) = \frac{1}{2} \left(\ln \frac{27}{16} + \pi \right)$	① ① 1 1	

2/3

MATHEMATICS Extension 2: Question 6...

Suggested Solutions

Marks

Marker's Comment

(c)(ii) let $t = \tan \frac{x}{2}$ when $x=0$ $t=0$
 when $x=\frac{\pi}{2}$ $t=1$

$$\therefore I = \int_0^1 \frac{1-t^2}{1+t^2} \times \frac{2dt}{1+t^2}$$

$$= 2 \int_0^1 \frac{1-t^2}{1+t^2+t^2+1-t^2} \frac{dt}{(1+t^2)}$$

$$= 2 \int_0^1 \frac{1-t^2}{(2+t^2)(1+t^2)} dt$$

$$= \int_0^1 \frac{1-t^2}{(2+t^2)(1+t^2)} dt$$

$$= \frac{1}{5} \times \frac{1}{2} \left(\pi + \ln \frac{27}{16} \right)$$

$$= \frac{1}{10} \left(\pi + \ln \frac{27}{16} \right)$$

(d) $u = \ln x$ $\frac{du}{dx} = \frac{1}{x}$
 $\frac{dv}{dx} = x^2$ $v = \frac{1}{3}x^3$

$$\int_1^3 x^2 \ln x dx = \left[\frac{1}{3} x^3 \ln x \right]_1^3 - \frac{1}{3} \int_1^3 x^2 dx$$

$$= (9 \ln 3 - 0) - \frac{1}{3} \left[\frac{1}{3} x^3 \right]_1^3$$

$$= 9 \ln 3 - \frac{1}{9} (27 - 1)$$

$$= 9 \ln 3 - \frac{26}{9}$$

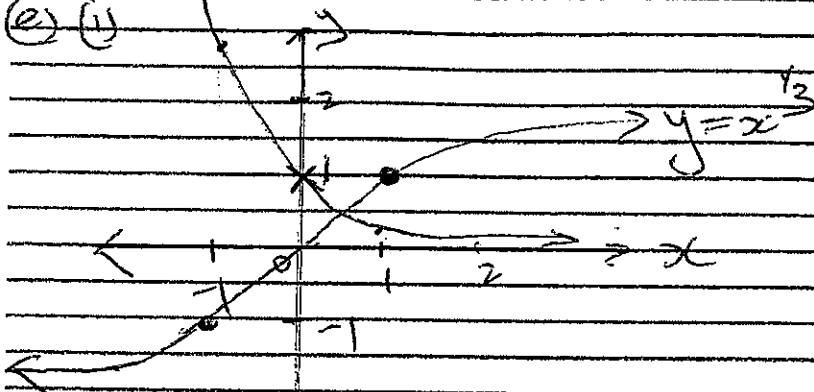
MATHEMATICS Extension 2: Question.....

Suggested Solutions

Marks

Marker's Comments

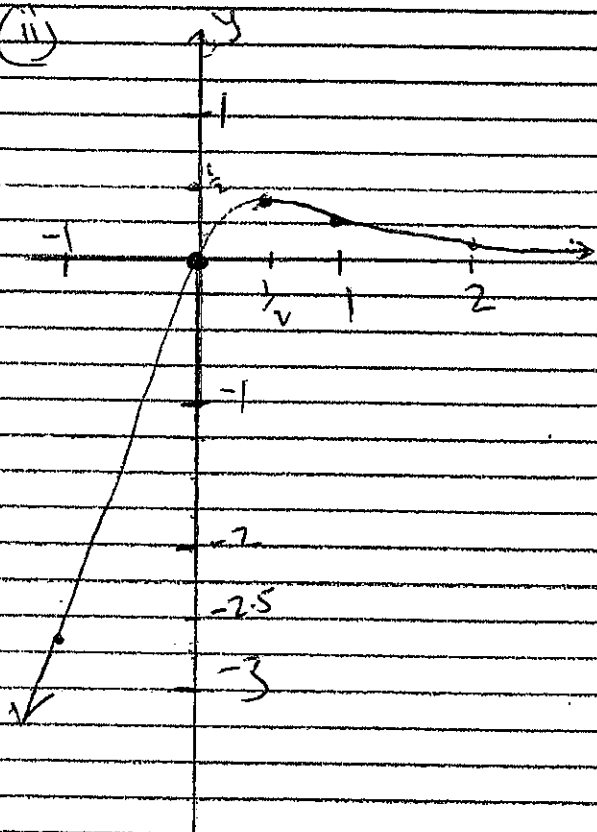
(i) (1)



*common error
was students
drew $y = x^3$
instead of $y = x^{1/3}$

*1 for each graph

(ii)



*no scale, lost a
mark

*poorly done.

$\frac{1}{2}$ for scale

$\frac{1}{2}$ for horizontal
asymptote for $x > 0$

$\frac{1}{2}$ for shape

$\frac{1}{2}$ for when $x < 0$

(iii) $m < 0$

1

$m < 0 \Rightarrow \frac{1}{2} \text{ mk}$

$m = 0 \Rightarrow 0 \text{ mk}$

$$a) e = \frac{1}{2}$$

$$ii) \frac{dy}{dx} = -\frac{3x}{4y}$$

$$\text{Eq of tangent } y - y_1 = -\frac{3x_1}{4y_1}(x - x_1)$$

$$4yy_1 - 4y_1^2 = -3xx_1 + 3x_1^2$$

$$\frac{xx_1}{4} + \frac{yy_1}{3} = \frac{x_1^2}{4} + \frac{y_1^2}{3}$$

Since $P(x_1, y_1)$ lies on ellipse $\frac{x_1^2}{4} + \frac{y_1^2}{3} = 1$

$$\therefore \text{Eq of tangent at } P: \frac{xx_1}{4} + \frac{yy_1}{3} = 1 \quad \#$$

$$iii) \text{ Directrix } x = \frac{a}{e} = \frac{2}{\frac{1}{2}} = 4$$

$$\text{Sub } x=4 \text{ into } \frac{xx_1}{4} + \frac{yy_1}{3} = 1$$

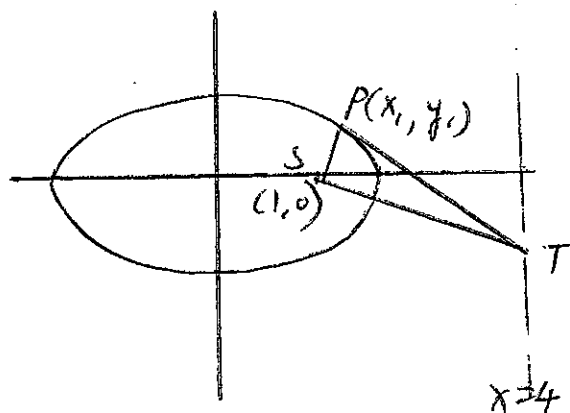
$$x_1 + \frac{yy_1}{3} = 1$$

$$y = \frac{3(1-x_1)}{y_1}$$

$$\therefore T = \left(4, \frac{3(1-x_1)}{y_1} \right)$$

$$\sim T = \left(-4, \frac{3(1+x_1)}{y_1} \right) \quad \text{optional}$$

iv)



$$m(ST) = \frac{3(1-x_1)}{y_1} - 0 / 4 - 1$$

$$h(ST) = \frac{(1-x_1)}{y_1}$$

1m well done

1m

1m

1m

1m
mention $P(x_1, y_1)$ lies
on ellipse

1m + 1m
well done

$$\begin{aligned} S &= (1, 0) \\ m(ST) &= \frac{1-x_1}{y_1} \quad \left. \begin{array}{l} \text{any 2} \\ \text{correct} \\ \text{get 1m} \end{array} \right\} \\ m(PS) &= \frac{y_1}{x_1-1} \quad \left. \begin{array}{l} \text{All 3} \\ \text{correct} \\ \text{get 2m} \end{array} \right\} \end{aligned}$$

$$(iv) m(PS) = \frac{y_1 - 0}{x_1 - 1} = \frac{y_1}{x_1 - 1}$$

$$m(ST) \times m(PS) = \frac{1-x_1}{y_1} \times \frac{y_1}{x_1-1} = -1$$

$$\therefore PS \perp ST \#$$

v) Eq of chord of contact from

$$R(x_0, y_0) \text{ is } \frac{xx_0}{4} + \frac{yy_0}{3} = 1 \#$$

$$vi) S(ae, 0) = (1, 0)$$

$$\frac{x_0-1}{4} + \frac{y_0-0}{3} = 1$$

$$\therefore x_0 = 4$$

$\therefore R$ lies on directrix #

$$b) \frac{x^2}{a^2} - \frac{(px+q)^2}{b^2} = 1$$

$$x^2(b^2 - a^2p^2) - 2(a^2pq)x - (a^2q^2 + a^2b^2) = 0 \quad 1m$$

For tangent, $\Delta = 0$

$$[-2(a^2pq)]^2 - 4(b^2 - a^2p^2)(a^2q^2 + a^2b^2) = 0 \quad 1m$$

$$4a^4p^2q^2 + 4a^2(b^2 - a^2p^2)(q^2 + b^2) = 0$$

$$\div 4a^2 \quad a^2p^2q^2 + b^2q^2 - a^2p^2q^2 + b^4 - a^2b^2p^2 = 0$$

$$b^2q^2 + b^4 - a^2b^2p^2 = 0$$

$$\div b^2 \quad q^2 + b^2 - a^2p^2 = 0$$

$$\therefore q^2 = a^2p^2 - b^2 \#$$

1 m

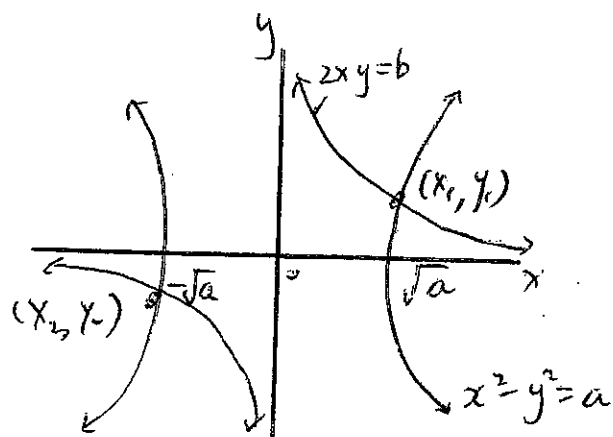
1 m well done

easy 2 m.

MANY judge.

1 m.

c) i)



ii) $z = x + iy$

$$z^2 = x^2 - y^2 + 2ixy = a + ib$$

$$\therefore x^2 - y^2 = a$$

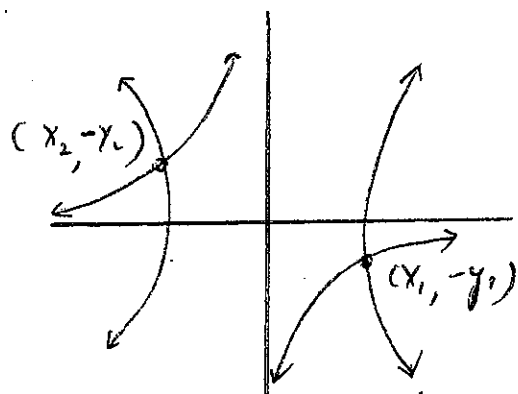
$$2xy = b$$

From graph i- (i), there are 2 distinct points of intersection

$\therefore$ 2 distinct square roots of $z = a + ib$

$$\text{i.e. } (x_1, y_1) = (x_2, y_2)$$

iii)



For $b < 0$, 2 distinct square roots of $z = a + ib$ will be

$$(x_1, -y_1) \text{ and } (x_2, -y_2)$$

$\therefore$ conjugates of the solns in part ii.

l.3

1m for each graph
must show x intercepts
 $\sqrt{a}$, $-\sqrt{a}$

many wrote $a, -a$
instead (-1m)

must show the
algebraic solutions
1m

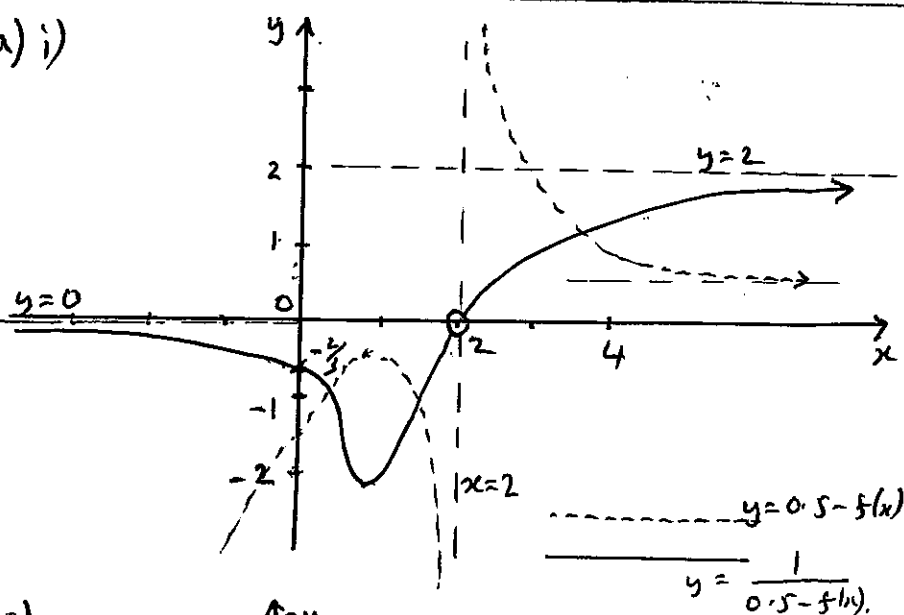
must mention
conjugates 1m

Suggested Solutions

Marks

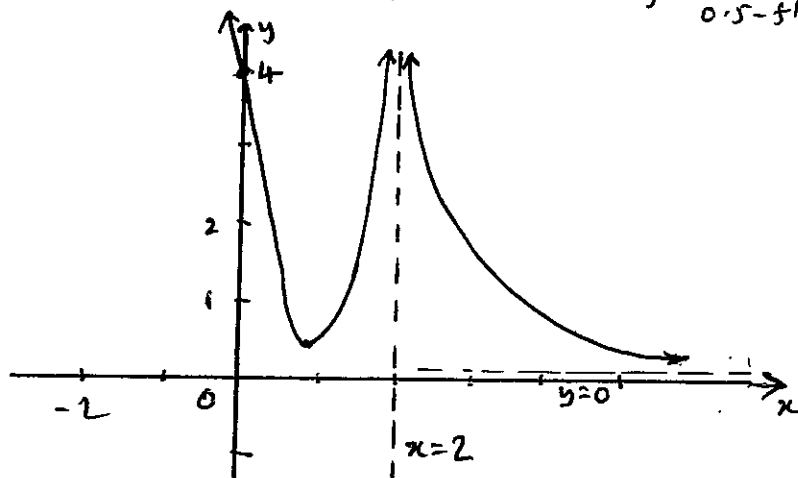
Marker's Comments

a) i)



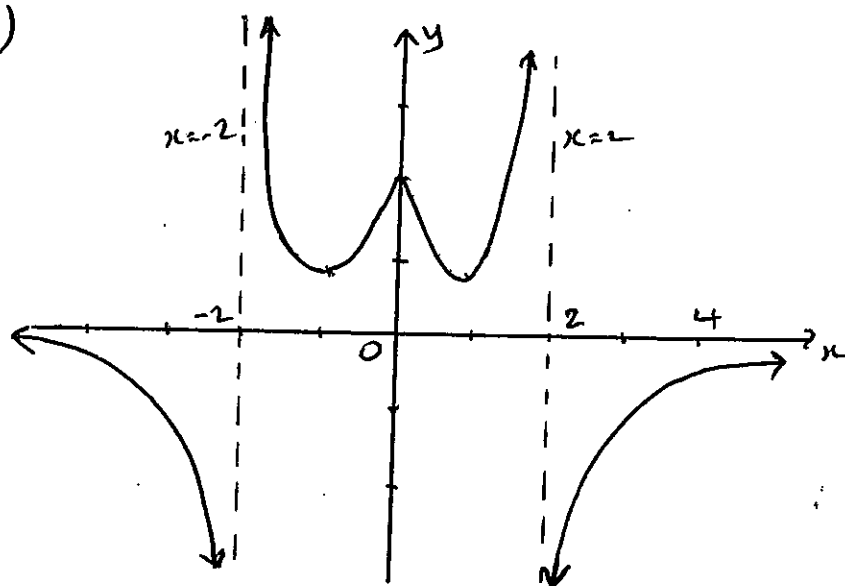
- 1 Asymptote $y=2$
- $\frac{1}{2}$ Asymptote $y=0$
- $\frac{1}{2}$ Intercept at $(0, -\frac{2}{3})$
- $\frac{1}{2}$ Hole at $(2,0)$
- $\frac{1}{2}$ Minimum near $(1,-2)$

ii)



- 1 Intercept at $(0, 4)$
- 1 Minimum near $(1, \frac{1}{2})$
(Some $\frac{1}{2}$ given here)
- Mark deducted if right hand section wrong.

iii)



- 1 for symmetry in $x=0$
- 1 for making the join "pointy".
(Some $\frac{1}{2}$ marks for indeterminate slash at $x=0$)

Suggested Solutions		Marks	Marker's Comments
iv)		1 1	for starting at $(-2, 0)$ for monotonic increasing. The intercept was indeterminate but would be somewhere near 7.
b) i)	$(1-\sqrt{x})^{n-1}\sqrt{x} = (1-\sqrt{x})^{n-1}(1+\sqrt{x}-1)$ $= (1-\sqrt{x})^{n-1} + (1-\sqrt{x})^{n-1}(\sqrt{x}-1)$ $= \underline{\underline{(1-\sqrt{x})^{n-1} - (1-\sqrt{x})^n}}$	2	Most people got two marks here. Need to be careful with given result
ii)	$I_n = \int_0^1 (1-\sqrt{x})^n dx$ $= \left[x(1-\sqrt{x})^n \right]_0^1 - \int_0^1 x n(1-\sqrt{x})^{n-1} \left(-\frac{1}{2\sqrt{x}} \right) dx$ (Integration by Parts) $= 0 + \frac{n}{2} \int_0^1 (1-\sqrt{x})^{n-1} \sqrt{x} dx$ $= \frac{n}{2} \int_0^1 (1-\sqrt{x})^{n-1} - (1-\sqrt{x})^n dx \quad (\text{Part (i)})$ $= \frac{n}{2} (I_{n-1} - I_n)$	1 1	for integration by parts successfully completed
	$\therefore I_n \left(1 + \frac{n}{2} \right) = \frac{n}{2} I_{n-1}$ $\therefore \underline{\underline{I_n = \frac{n I_{n-1}}{n+2}}}$	1	for use of (i) after integration If used before, only $\frac{1}{2}$ given for that by itself. manipulation and answer.

Suggested Solutions

Marks

Marker's Comments

iii) $I_1 = \int_0^1 1 - \sqrt{x} \, dx$

$$= \left[x - \frac{2x^{3/2}}{3} \right]_0^1 = \left(1 - \frac{2}{3} \right) - 0 = \frac{1}{3}.$$

$$\therefore I_2 = \frac{2}{4} I_1 = \frac{1}{6}$$

$$I_3 = \frac{3}{5} I_2 = \frac{3}{5} \times \frac{1}{6} = \underline{\underline{\frac{1}{10}}}$$

c) $\int_{-\pi/2}^{\pi/2} \sin^5 \theta \, d\theta = 0$ (Integral of an odd function between symmetric limits)

d) i) Using result $|z|^2 = z\bar{z}$

$$|z+w|^2 + |z-w|^2 = (z+w)(\bar{z}+\bar{w}) + (z-w)(\bar{z}-\bar{w})$$

$$= (z+w)(\bar{z}+\bar{w}) + (z-w)(\bar{z}-\bar{w})$$

$$= z\bar{z} + w\bar{z} + z\bar{w} + w\bar{w} + z\bar{z} - z\bar{w} - w\bar{z} + w\bar{w}$$

$$= 2(z\bar{z} + w\bar{w})$$

$$= \underline{\underline{2(|z|^2 + |w|^2)}}$$

ii) (These numbers can be represented by the diagonals and sides of a parallelogram.)

The sum of the squares of the diagonals in a parallelogram equals the sum of the squares of the four sides.

1

for $I_1 = \frac{1}{3}$
or $I_0 = 1$

1

for manipulation.

1

No mark if no reason at all
Very few people mentioned symmetric limits

←

2

In a given result it is important to show all steps.

This line (or equivalent) is the critical line.
Several people lost a mark for not including it.
Many people used $x+iy$ forms.

1

Some $\frac{1}{2}$ marks given for 'typos'.

Suggested Solutions

Marks

Marker's Comments

Question 9

a) (i) T is $(a \cos \theta, a \sin \theta)$

$$x^2 + y^2 = a^2$$

$$2x + 2y \frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = -\frac{x}{y}$$

$$\text{at } (a \cos \theta, a \sin \theta), \frac{dy}{dx} = \frac{-a \cos \theta}{a \sin \theta} = -\frac{\cos \theta}{\sin \theta}$$

Equation of TM is

$$y - a \sin \theta = -\frac{\cos \theta}{\sin \theta} (x - a \cos \theta)$$

$$y \sin \theta - a \sin^2 \theta = -x \cos \theta + a \cos^2 \theta$$

$$x \cos \theta + y \sin \theta = a \quad \text{since } \sin^2 \theta + \cos^2 \theta = 1$$

$$\text{When } y = 0, x = \frac{a}{\cos \theta} \Rightarrow M \left(\frac{a}{\cos \theta}, 0 \right)$$

(ii) substitute $x = a \sec \theta$ into

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

$$\sec^2 \theta - \frac{y^2}{b^2} = 1$$

$$y^2 = b^2 (\sec^2 \theta - 1)$$

$$y = \pm b \tan \theta$$

P is $(a \sec \theta, b \tan \theta)$

(iii) Q $(a \sec \theta, b \tan \theta)$ but $\theta = \frac{\pi}{2} - \phi$

Q is $(a \sec(\frac{\pi}{2} - \phi), b \tan(\frac{\pi}{2} - \phi))$

i.e. $(a \csc \phi, b \cot \phi)$

$$m_{PQ} = \frac{b \cot \phi - b \tan \theta}{a \csc \phi - a \sec \theta}$$

$$= \frac{b}{a} \left\{ \frac{\cos \phi - \sin \theta}{\sin \phi \cos \theta} \right\} \quad \begin{matrix} \times \text{ above + below} \\ \left\{ \frac{1}{\sin \phi} - \frac{1}{\cos \theta} \right\} \text{ by } \sin \phi \cos \theta \end{matrix}$$

$$m_{PQ} = \frac{b}{a} \left\{ \frac{\cos^2 \theta - \sin^2 \theta}{\cos \theta - \sin \theta} \right\}$$

$$= \frac{b}{a} \{ \cos \theta + \sin \theta \}$$

Eqn. of PQ is

$$y - b \tan \theta = \frac{b}{a} \{ \cos \theta + \sin \theta \} (x - a \sec \theta)$$

$$ay - ab \sin \theta = bx \cos \theta - ab + bx \sin \theta - ab \sin \theta$$

$$\underline{ay = (\cos \theta + \sin \theta) bx - ab}$$

Gradient

Equation (any form)

$$x = \frac{a}{\cos \theta}, y = 0$$

proof

Gradient in terms of θ

Straight line formula with some simplification

answer

Suggested Solutions

Marks

Marker's Comments

c) Let $P(n)$ be the proposition that

$$\tan^{-1}\left(\frac{1}{2 \times 1}\right) + \tan^{-1}\left(\frac{1}{2 \times 2}\right) + \dots + \tan^{-1}\left(\frac{1}{2 \times n}\right) = \frac{\pi}{4} - \tan^{-1}\left(\frac{1}{2n+1}\right)$$

test $n=1$

$$\text{LHS} = \tan^{-1}\left(\frac{1}{2}\right) \quad \text{RHS} = \frac{\pi}{4} - \tan^{-1}\left(\frac{1}{3}\right) \quad (1)$$

$$\text{as } \tan^{-1}x + \tan^{-1}y = \tan^{-1}\left(\frac{x+y}{1-xy}\right) \quad 0 \leq x \leq 1 \quad 0 \leq y \leq 1$$

$$\text{consider } \tan^{-1}\left(\frac{1}{2}\right) + \tan^{-1}\left(\frac{1}{3}\right) = \tan^{-1}\left[\frac{\frac{1}{2} + \frac{1}{3}}{1 - \frac{1}{2} \times \frac{1}{3}}\right] = \frac{\frac{5}{6}}{\frac{5}{6}} = 1$$

$$= \frac{\pi}{4} \quad (1)$$

$$\therefore \tan^{-1}\left(\frac{1}{2}\right) = \frac{\pi}{4} - \tan^{-1}\left(\frac{1}{3}\right) \therefore P(1) \text{ is true.}$$

Assume $P(k)$ is true for $k \in \mathbb{Z}^+$

$$\text{i.e. } \tan^{-1}\left(\frac{1}{2 \times 1}\right) + \tan^{-1}\left(\frac{1}{2 \times 2}\right) + \dots + \tan^{-1}\left(\frac{1}{2 \times k}\right) = \frac{\pi}{4} - \tan^{-1}\left(\frac{1}{2k+1}\right)$$

To prove $P(k+1)$ is true.

$$\text{i.e. } \tan^{-1}\left(\frac{1}{2 \times 1}\right) + \tan^{-1}\left(\frac{1}{2 \times 2}\right) + \dots + \tan^{-1}\left(\frac{1}{2 \times k}\right) + \tan^{-1}\left(\frac{1}{2 \times (k+1)}\right)$$

$$= \frac{\pi}{4} - \tan^{-1}\left(\frac{1}{2(k+1)}\right)$$

$$= \frac{\pi}{4} - \tan^{-1}\left(\frac{1}{2k+2}\right)$$

$$\therefore \text{LHS} = \tan^{-1}\left(\frac{1}{2 \times 1}\right) + \tan^{-1}\left(\frac{1}{2 \times 2}\right) + \dots + \tan^{-1}\left(\frac{1}{2 \times k}\right) + \tan^{-1}\left(\frac{1}{2 \times (k+1)}\right)$$

$$= \frac{\pi}{4} - \tan^{-1}\left(\frac{1}{2(k+1)}\right) + \tan^{-1}\left(\frac{1}{2(k+1)}\right) \text{ by assumption}$$

$$\text{consider } \tan^{-1}\left(\frac{1}{2(k+1)}\right) + \tan^{-1}\left(\frac{1}{2(k+1)}\right) \quad (1)$$

$$= \tan^{-1}\left[\frac{\frac{1}{2(k+1)} + \frac{1}{2(k+1)}}{1 - \frac{1}{2(k+1)} \times \frac{1}{2(k+1)}}\right] \text{ from above.}$$

$$= \tan^{-1}\left[\frac{(2k+3) + 2(k+1)}{2(k+1)^2 - 1}\right] \quad (1)$$

$$= \tan^{-1}\left[\frac{2k+3+2k^2+4k+2}{(2k^2+4k+2)(2k+3)-1}\right]$$

$$= \tan^{-1}\left[\frac{2k^2+6k+5}{4k^2+14k+5}\right]$$

$$= \tan^{-1}\left[\frac{2k^2+6k+5}{(2k^2+6k+5)(2k+1)}\right]$$

$$\tan^{-1}\left(\frac{1}{2k+1}\right)$$

$$\therefore \tan^{-1}\left(\frac{1}{2(k+1)}\right) + \tan^{-1}\left(\frac{1}{2(k+1)}\right) = \tan^{-1}\left(\frac{1}{2k+1}\right)$$

$$- \tan^{-1}\left(\frac{1}{2(k+1)}\right) + \tan^{-1}\left(\frac{1}{2(k+1)}\right) = -\tan^{-1}\left(\frac{1}{2k+3}\right)$$

$$\frac{\pi}{4} - \tan^{-1}\left(\frac{1}{2(k+1)}\right) + \tan^{-1}\left(\frac{1}{2(k+1)}\right) = \frac{\pi}{4} - \tan^{-1}\left(\frac{1}{2k+3}\right)$$

$$\therefore P(k+1) \text{ is true.} \quad (1)$$

$\therefore P(n)$ is true by principle of mathematical induction

(1)

(1)

Must read question and not change given information

Note: $0 \leq y \leq 1$
 $0 \leq x \leq 1$

Formula was given for $\tan^{-1}x + \tan^{-1}y$
only (1) if $y = -\frac{1}{3}$ used.

(1) Assumption line

(1) using formula

No loss of mark if "-" used.

correct derivation of final answer (complete proof).

Suggested Solutions	Marks	Marker's Comments
(iv) y coordinate for PO is always $y = -b$ $\therefore$ The required point is $(0, -b)$ ①	①	$(0, -b)$.
(v) as $\theta \rightarrow \frac{\pi}{2}$ becomes $ay \rightarrow (\cos\theta + \sin\theta)bx - ab$ $ay \rightarrow (0+1)bx - ab$ $ay \rightarrow bx - ab$ $y \rightarrow \frac{b}{a}x - b$ ① $\therefore$ PO is parallel to the asymptote $y = \frac{bx}{a}$ ①	①	$y \rightarrow \frac{b}{a}x - b$
b) (i) $f(x) = \frac{1}{1+\tan x}$ $f(x) + f(\frac{\pi}{2}-x) = \frac{1}{1+\tan x} + \frac{1}{1+\tan(\frac{\pi}{2}-x)}$ $= \frac{1}{1+\tan x} + \frac{1}{1+\cot x}$ ① $= \frac{1}{1+\tan x} + \frac{\tan x}{\tan x + 1}$ $= \frac{1+\tan x}{1+\tan x}$ ① $= 1$	①	Equation of asymptote.
b) (ii) $\int_a^a f(x) dx = \int_0^a f(a-x) dx$ $\therefore \int_0^{\frac{\pi}{2}} \frac{1}{1+\tan x} dx = \int_0^{\frac{\pi}{2}} \frac{1}{1+\tan(\frac{\pi}{2}-x)} dx$ ① But $\frac{1}{1+\tan x} + \frac{1}{1+\tan(\frac{\pi}{2}-x)} = 1$ $\therefore \int_0^{\frac{\pi}{2}} \frac{1}{1+\tan x} dx = \int_0^{\frac{\pi}{2}} (1 - \frac{1}{1+\tan x}) dx$ $2 \int_0^{\frac{\pi}{2}} \frac{1}{1+\tan x} dx = \int_0^{\frac{\pi}{2}} 1 dx$ ① $2 \int_0^{\frac{\pi}{2}} \frac{1}{1+\tan x} dx = [x]_0^{\frac{\pi}{2}}$ $\therefore \int_0^{\frac{\pi}{2}} \frac{1}{1+\tan x} dx = \frac{1}{2} (\frac{\pi}{2} - 0)$ $= \frac{\pi}{4}$ ①	①	changing $\tan(\frac{\pi}{2}-x)$.
	①	answer
	①	Quoting or proving shift formula.
	①	Showing 2I
	①	Answer